

Candidate Name: _____

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MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2015
Higher 2

H2 BIOLOGY

Paper 1 Multiple Choice Questions

Additional Materials: Multiple Choice Answer Sheet

9648/01**23 September 2015****1 hour 15 minutes**

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write in soft pencils.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Write your name, civics group and index number on the Multiple Choice Answer Sheet provided.

There are **forty** questions on this paper. Answer **all** questions. For each question, there are four possible answers **A, B, C** and **D**. Choose the one you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

You may keep this booklet after the examination.

Answers

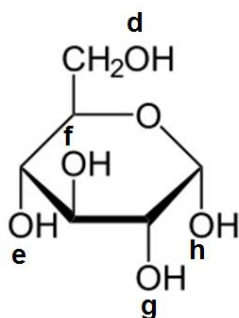
This paper consists of __ printed pages.

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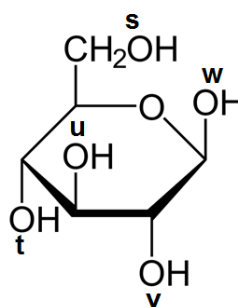
Question 1

The diagram shows two isomers of an hexose labelled I and II.

Four possible bonding positions are labelled **d**, **e**, **f**, **g** and **h** on one isomer, and **s**, **t**, **u**, **v** and **w** on another isomer.



Isomer I



Isomer II

Which isomer and bonding positions are involved in the formation of glycogen?

	Repeating units	Bonding position
A.	Isomer I	d-g and d-h
B.	Isomer I	e-h and d-h
C.	Isomer II	s-v and s-w
D.	Isomer II	t-w and s-w

Question 2

Which of the following statements about the Golgi apparatus are correct?

1. It may assist in the aerobic oxidation of glucose, especially in actively respiring tissues like muscles.
2. It forms lysosomes, which are involved in cell autolysis.
3. It is made of membrane-bound sacs called cisternae
4. It is involved lipid synthesis and the packaging of proteins prior to their secretion from the cell.
5. It is commonly found in animal cells, especially those with a secretory function, but it is never found in plant cells.
6. It forms vesicles that migrate along microtubules and fuse to become a cell plate in plant cells.

A. 1, 3 and 4

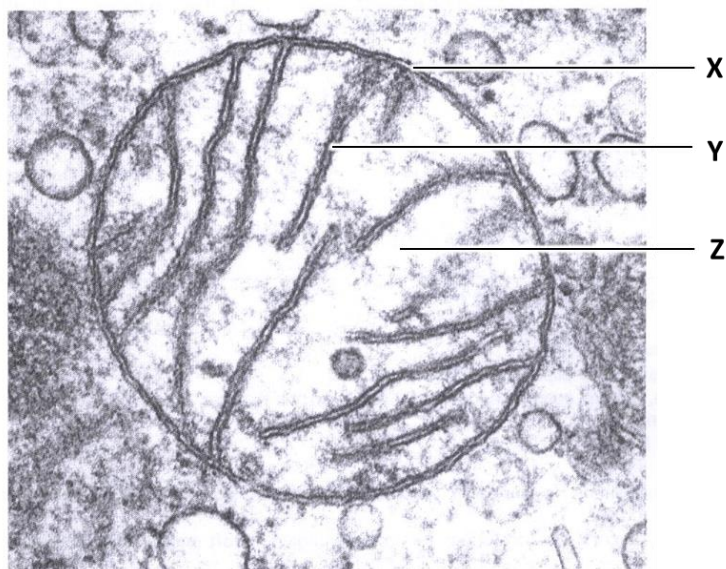
B. 2, 3 and 6

C. 2, 4 and 5

D. 1, 5 and 6

Question 3

The electron micrograph below shows an organelle.



Which of the following statements about the labelled structures are correct?

1. X serves to compartmentalise the organelle from the extracellular environment.
2. X is made up of a phospholipid bilayer and it is selectively permeable to substances.
3. Y has embedded photosystems that are involved in the absorption of light.
4. Y has embedded ATP synthase that synthesizes ATP.
5. Z contains DNA that has histones that help to package DNA present in the organelle.
6. Z contains glycolytic enzymes that oxidises glucose to pyruvate.

A. 2 and 4

B. 5 and 6

C. 1, 2 and 4

D. 1, 3 and 6

Question 4

Which of the following statements about enzymes are false?

1. All enzymes are globular proteins.
2. Enzymes catalyse reactions by decreasing the activation energy.
3. A prosthetic group is tightly bound to the enzyme, while a coenzyme is a loosely bound to the enzyme.
4. The effect of competitive inhibitors can be reduced by increasing substrate concentration.
5. An allosteric binding site refers to the active site that has undergone an induced fit.

A. 1 and 3

B. 1 and 5

C. 2 and 4

D. 3 and 5

Question 5

A solution of amylase is saturated with starch at pH6.5 at 30°C. The best method to obtain a faster yield of products would be to

- A. reduce the amount of substrate so that there are more free proteases to catalyse the reaction.
- B. increase the pH so that the proteases are more active.
- C. double the temperature to double the rate of reaction.
- D. double the concentration of amylase to double the rate of reaction.

Question 6

What would be the ploidy of the gametes generated if non-disjunction occurs at first meiotic division?

- A. both (n+1) and n
- B. both (n+1) and (n-1)
- C. either (n+1) or (n-1)
- D. (n+1), n, and (n-1)

Question 7

Which of the following statements correctly describes the differences between DNA replication in a plant cell and that in *E.coli* bacteria?

1. *E. coli* replicates chromosomal DNA using the rolling circle mechanism of replication, unlike plant cells.
2. The plant cell has multiple origin of replication while *E.coli* has only one origin of replication.
3. Plant cell replication requires a DNA primer while *E.coli* replication requires a RNA primer.
4. *E.coli* DNA polymerase has no 3' → 5' exonuclease activity, unlike the plant cell.
5. Production of okazaki fragments in the plant cell, which are absent in *E. coli*.

- A. 2 only
- B. 4 only
- C. 1, 3 and 4
- D. 2, 4 and 5

Question 8

In a laboratory of Molecular Biology, the amino acids sequence of an armadillo intestine protein has been partially determined. The tRNA molecules used in the synthesis have the following anticodons:

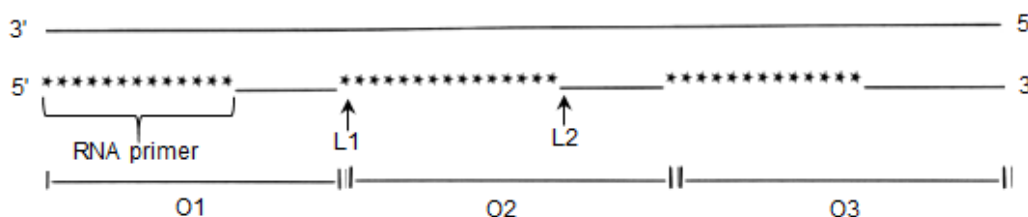
3' UAC 5' 3' CGA 5' 3' GGA 5' 3' GCU 5' 3' UUU 5' 3' GGA 5'

What is the DNA nucleotide sequence of the non-template DNA strand?

- A. 5' ATG-GCT-GGT-CGA-AAA-CCT 3'
- B. 5' ATG-GCT-CCT-CGA-AAA-CCT 3'
- C. 5' ATG-GCT-GCT-CGA-AAA-GCT 3'
- D. 5' ATG-GGT-CCT-CGA-AAA-CGT 3'

Question 9

The diagram shows a DNA template with the lagging strand prior to the removal of the RNA primers.



Which row correctly shows the events taking place during the synthesis of the lagging strand?

	first Okazaki fragment synthesised	site of phosphodiester bond formation catalysed by DNA ligase
A.	O1	L1
B.	O1	L2
C.	O3	L1
D.	O3	L2

Question 10

Which of the following mutations is most likely to cause a phenotypic change?

- A. a duplication of all introns
- B. a single nucleotide deletion in an exon coding for an active site
- C. a nucleotide substitution in an exon coding for a transmembrane domain
- D. a frameshift mutation one triplet away from the 3' end of the non-template strand

Question 11

Which of the following is not true about cancer cells?

- A. Metastasis establishes new tumours distant from the site of the primary tumour.
- B. Angiogenesis forms new blood vessels and brings nutrients and oxygen to the tumour.
- C. Telomerase maintains the length of telomeres.
- D. They exhibit contact inhibition.

Question 12

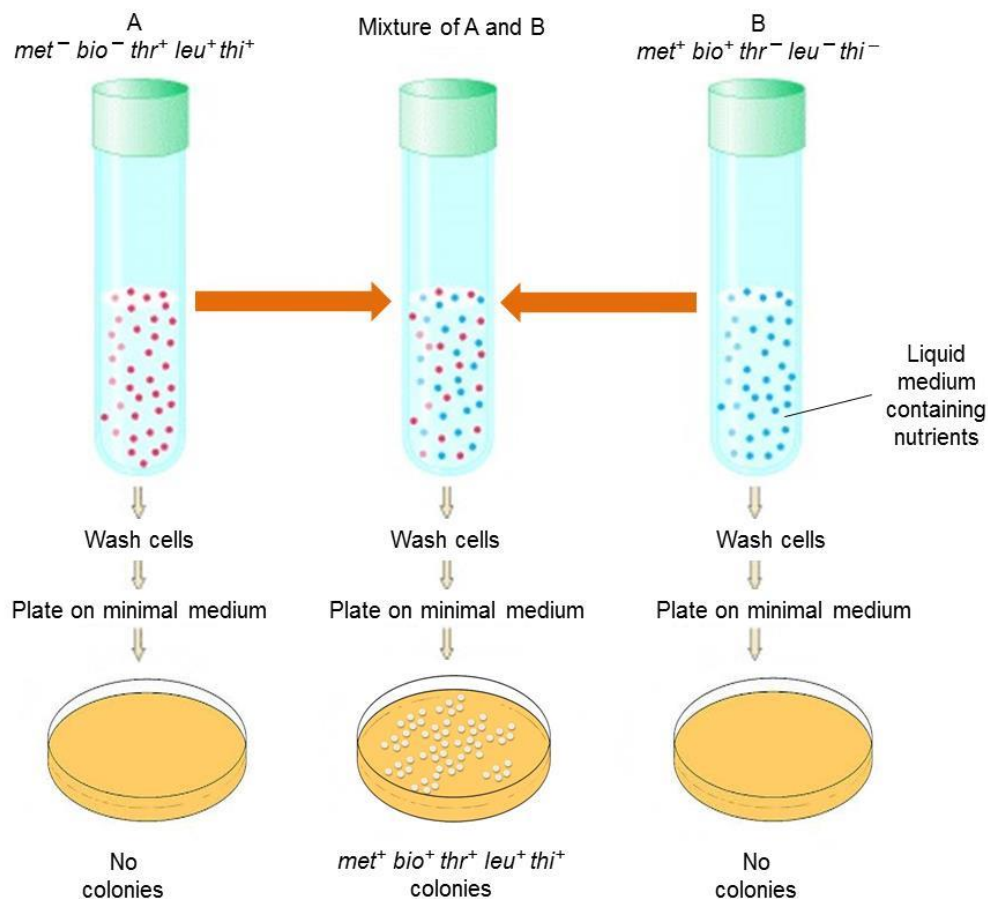
Which of the following statements is true about bacterial plasmids?

- A. Plasmids replicate along with the chromosomal DNA by using chromosomal DNA as a template.
- B. Antibiotic-resistant genes on plasmids allow them to survive in antibiotic culture medium.
- C. Plasmid DNA strands run in opposite directions and the copy number varies with the type of plasmid.
- D. Binary fission ensures that plasmids are equally separated into daughter cells.

Question 13

In an experiment to study genetic recombination, two strains of *Escherichia coli* with different nutritional requirements were used. Strain A ($met^- bio^- thr^+ leu^+ thi^+$) would grow on a minimal medium only if the medium were supplemented with methionine and biotin, while strain B ($met^+ bio^+ thr^- leu^- thi^-$) would grow on a minimal medium only if it were supplemented with threonine, leucine and thiamine.

Some of the dishes were plated only with strain A bacteria, some only with strain B bacteria, and some with a mixture of strain A and strain B bacteria that had been incubated together for several hours in a liquid medium containing all required nutrients.

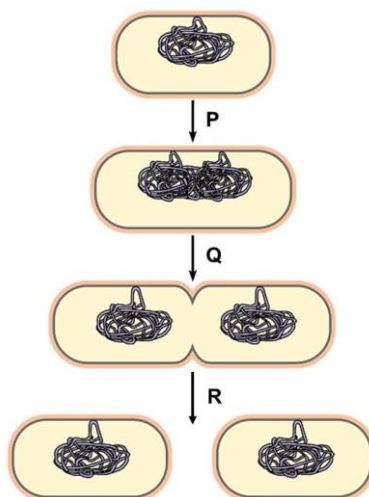


Which conclusion can be best drawn from the experiment?

- A. The $thr^+ leu^+ thi^+$ genes in strain A are located in its F plasmid.
- B. Cell-to-cell contact is necessary for the exchange of alleles between strain A and strain B.
- C. All the five genes are located on the F plasmid.
- D. Mutations resulted in the presence of $thr^+ leu^+ thi^+$ genes in strain B.

Question 14

Shown below are stages of bacterial reproduction.

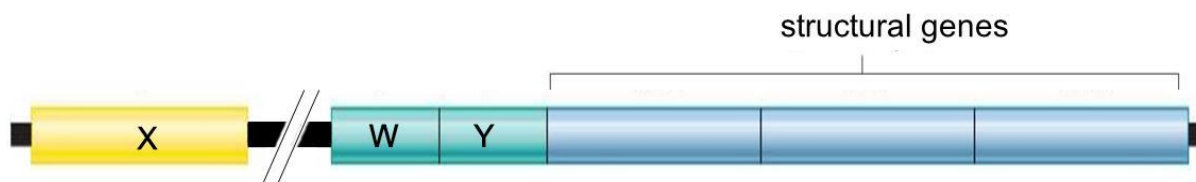


Which of the following statements is true?

- A. P can be blocked by introducing an inhibitor for RNA polymerase.
- B. P ensures genetic stability of the daughter cells.
- C. Q is the anaphase of the cell cycle, where chromosomes are pulled to opposite poles of the cell.
- D. R is cytokinesis, where the cytoplasm divides to produce two daughter cells.

Question 15

The diagram shows a length of DNA responsible for metabolising lactose in prokaryotes. A mutation occurred in X such that its protein product became non-functional.

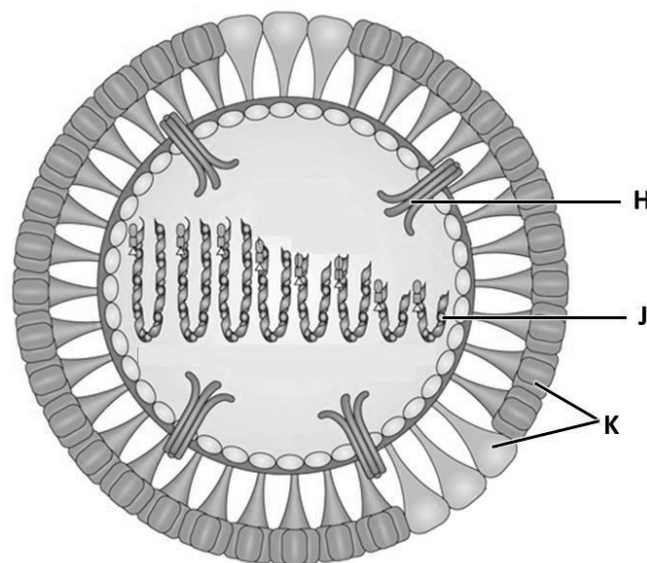


What is a possible outcome of the mutation if the bacteria cell is grown in a culture medium containing only lactose?

	Sequence bound by protein	Positive control of gene regulation	Negative control of gene regulation	Transcription of structural genes
A.	W	absent	present	no
B.	Y	absent	present	no
C.	W	present	absent	yes
D.	Y	present	absent	yes

Question 16

The structure of an influenza virus is shown below.



Which of the following statements are true?

1. Antigenic drift may result in changes in **K**.
2. **H** is derived from the host cell during viral release.
3. **J** can be directly used for translation to produce **H** and **K**
4. Viruses may be classified according to the absence or presence of **K**.

- A. 1 and 2
B. 1 and 3
C. 2 and 4
D. 3 and 4

Question 17

Telomerase RNA plays an important role in maintaining telomeres. The folding of telomerase RNA into its precise 3D conformation is an example of:

- A. transcriptional control
B. post-transcriptional control
C. translational control
D. post-translational control

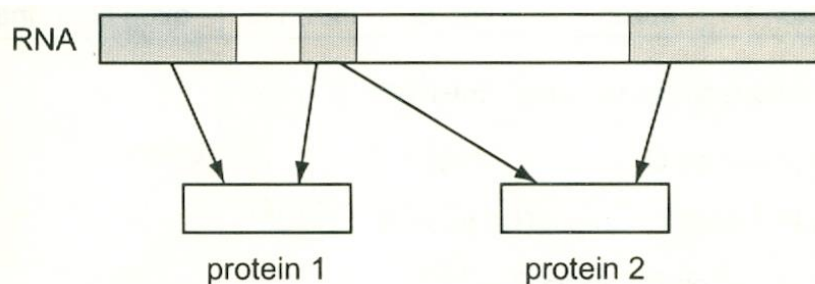
Question 18

Which statement regarding telomerase is false?

- A. It extends the 3' end of the daughter strands during DNA replication
B. It is a potential target for chemotherapy
C. It can be rendered non-functional by both proteases and nucleases
D. The telomerase gene is found in heterochromatin regions in most human adult cells

Question 19

RNA transcribed from a length of DNA of a chromosome was found to code for two different proteins that function as enzymes, as shown in the diagram.



Which of the following best describes the two proteins?

- A. Protein 1 and protein 2 are a result of differential gene expression.
- B. Protein 1 and protein 2 are a result of the cleavage of a polyprotein.
- C. Protein 1 and protein 2 function sequentially in the same metabolic pathway.
- D. Protein 1 and protein 2 usually perform similar functions in different cell types.

Question 20

A cross between two plants produced phenotypes in these numbers:

Long stem, pink flowers	186
Short stem, pink flowers	67
Short stem, white flowers	63
Long stem, white flowers	184

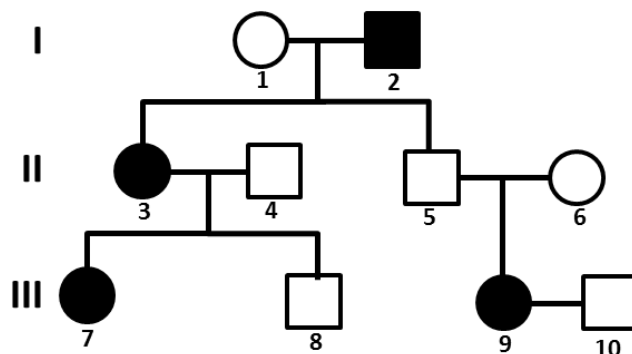
Assuming that gene **A/a** codes for flower colour and gene **B/b** codes for stem length, which of the following is true about the parent plants and the mode of inheritance of these traits?

	Phenotypes	Genotypes	Mode of inheritance
A.	Red flowers, long stem X White flowers, long stem	AaBb X aaBb	Epistasis Co-dominance
B.	Pink flowers, long stem X White flowers, long stem	AaBb X AaBb	Epistasis Complete dominance
C.	Pink flowers, long stem X White flowers, long stem	AaBb X aaBb	Unlinked genes Complete dominance
D.	Pink flowers, long stem X White flowers, short stem	AaBb X aabb	Unlinked genes Complete dominance

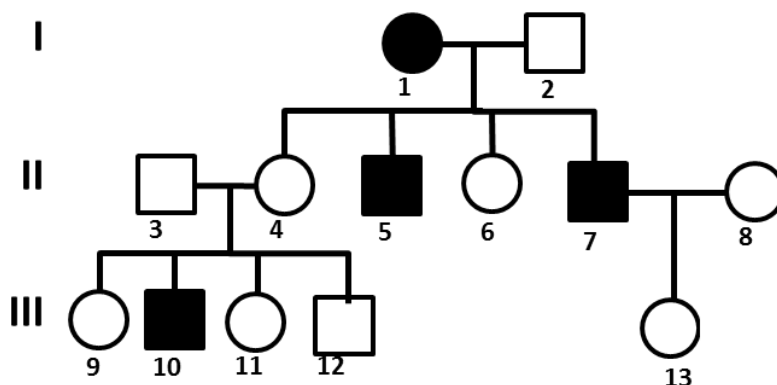
Question 21

Two pedigree charts showing the inheritance of different diseases are shown below.

Pedigree 1:



Pedigree 2:



Which of the following statements about the pedigree charts are false?

1. Pedigree 1 illustrates a sex-linked recessive disease as individual II-5 and II-6 produce affected individuals.
2. Pedigree 2 illustrates sex-linked dominant disease as affected mother I-1 produces affected sons.
3. A cross between individual III-8 and III-9 from pedigree 1 is likely to produce both affected and unaffected individuals.
4. A cross between individual III-13 from pedigree 2 with an unaffected male will produce all unaffected females and all affected sons.

- A. 1 and 2
B. 2 and 3
C. 2, 3 and 4
D. 1, 2 and 4

Question 22

Drosophila with disarranged thoracic bristles and wings that are held vertically upwards are said to be “deranged”. Crosses between pure-breeding deranged females and normal males, result in a 1:1 ratio of normal females to deranged males among the progeny.

What type of inheritance does this show?

- A. The gene for deranged is autosomal and dominant.
- B. The gene for deranged is autosomal and recessive.
- C. The gene for deranged is X-linked and dominant.
- D. The gene for deranged is X-linked and recessive.

Question 23

There are two hypotheses to explain the production of white, pale pink or dark pink flowers in a species of plant.

Hypothesis 1 There are two codominant alleles.

Hypothesis 2 There are three alleles, one for each flower colour.

Which procedure is the best way of testing these hypotheses?

- A. Analysis of the flower pigments in several different flowers by chromatography to find whether some plants contain more than one pigment.
- B. Controlled cross pollination of all the different colour varieties available, in all possible combinations, and recording the colours shown by the offspring.
- C. Controlled self-pollination of several individuals of each of the colour varieties and recording the colours shown by the offspring of each individual plant sampled.
- D. Surveying large wild populations and finding the ratios of the different colours in these.

Question 24

Which of the following statements regarding quantitative inheritance of phenotypes are false?

1. The environment plays an important role in quantitative inheritance.
2. Different genes with multiple alleles do not contribute to quantitative variation.
3. Identifying quantitative trait loci is relatively straightforward.
4. Quantitative inheritance is also known as continuous variation.
5. Different alleles at a single gene locus have large effects on the phenotype.

A. 2, 3 and 5

B. 1, 2 and 3

C. 3 and 4

D. 2 and 5

Question 25

Two different strains of *Drosophila*, strain A and B, each have a recessive mutation on separate genes that result in abnormally bright red eyes in both strains. Wild type flies have brown eyes. When a homozygous strain A fly is crossed with a homozygous strain B fly, all the progeny have dominant wild type eye colour.

The F1 progeny were allowed to breed among themselves.

What is the phenotypic ratio expected among their offspring?

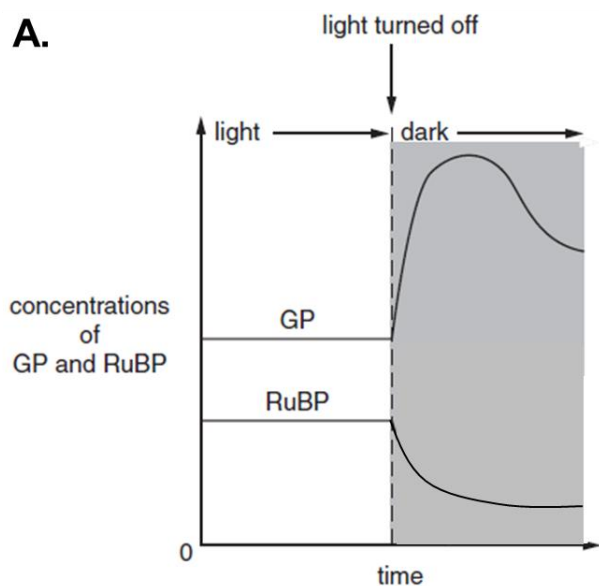
- A. 1 wild type :1 bright red eyes
- B. 3 wild type :1 bright red eyes
- C. 9 wild type: 7 bright red eyes
- D. 9 wild type : 6 dull red eyes : 1 bright red eyes

Question 26

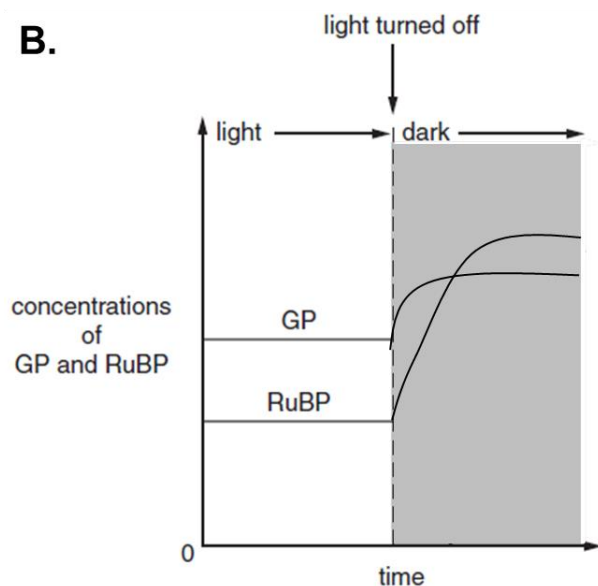
Concentrations of glycerate-3-phosphate (GP) and ribulose biphosphate (RuBP) were measured from samples of actively photosynthesising green algae in an experimental chamber.

Which of the following graphs show how the concentration of these compounds changes when the light source was turned off?

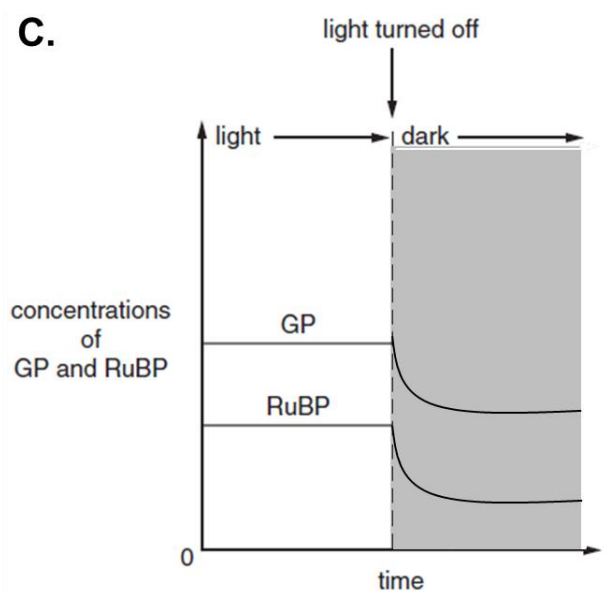
A.



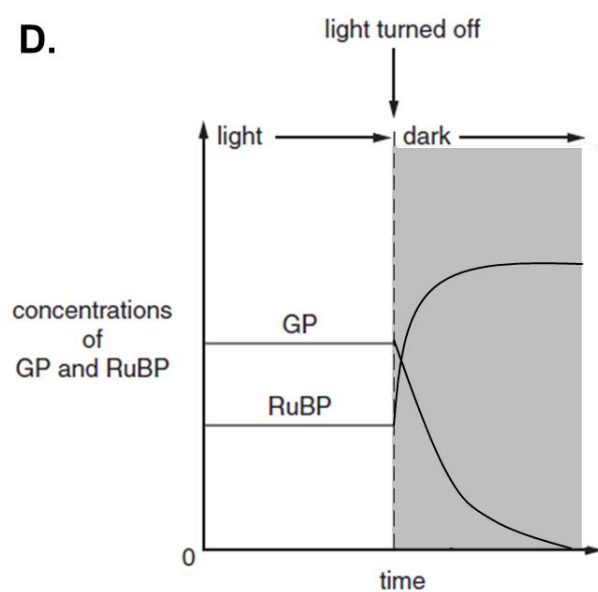
B.



C.

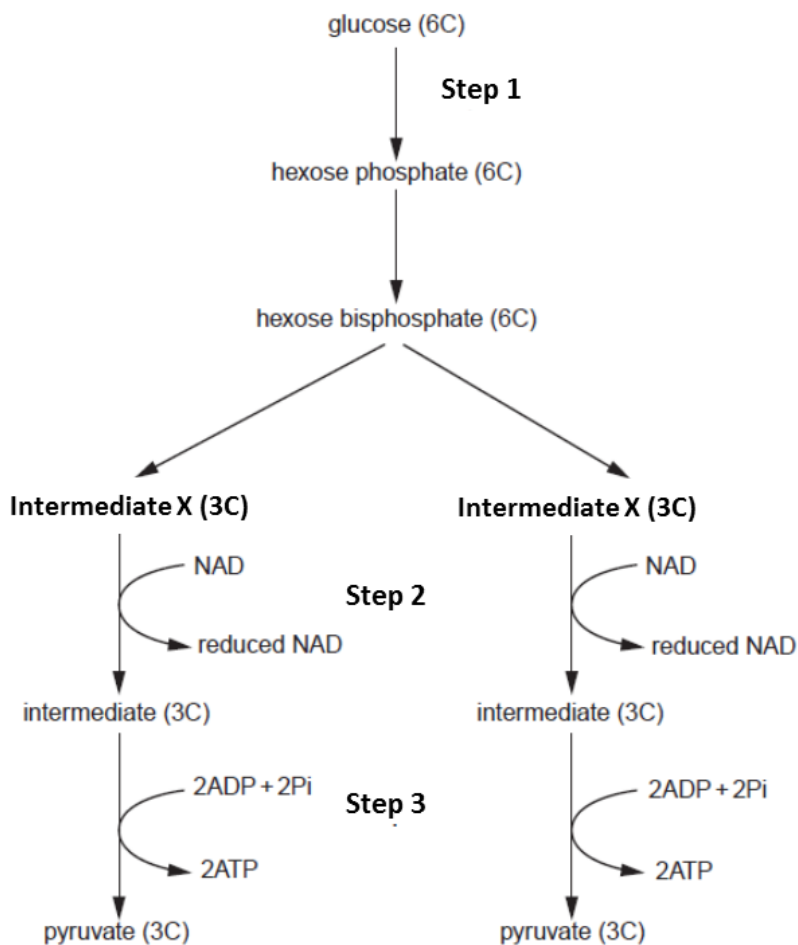


D.



Question 27

During glycolysis, glucose is converted by a series of steps into two molecules of pyruvate.



Which of the following correctly labels intermediate X, steps 1 – 3 and the location where step 3 occurs at?

	Location	Intermediate X	Step 1	Step 2	Step 3
A.	Cytosol	Glyceraldehyde -3-phosphate	Reduction	Oxidation	Oxidative phosphorylation
B.	Cytosol	Triose phosphate	Phosphorylation	Oxidation	Substrate level phosphorylation
C.	Mitochondrial matrix	Glycerate-3-phosphate	Reduction	Reduction	Substrate level phosphorylation
D.	Cytosol	Triose phosphate	Phosphorylation	Reduction	Substrate level phosphorylation

Question 28

The following statements describe respiration.

1. Two turns of the citric acid cycle is required to oxidize 1 molecule of glucose.
2. Four molecules of carbon dioxide are generated for every molecule of acetyl CoA introduced into the Krebs Cycle.
3. During aerobic respiration, glucose produces pyruvate, CO₂ and ATP in the cytoplasm of a muscle cell.
4. Aerobic respiration can produce about 19 times the amount of ATP produced in anaerobic respiration.

Which of the above statements is/are true?

- A. 1 only
B. 1 and 4
C. 2 and 4
D. 1, 3 and 4

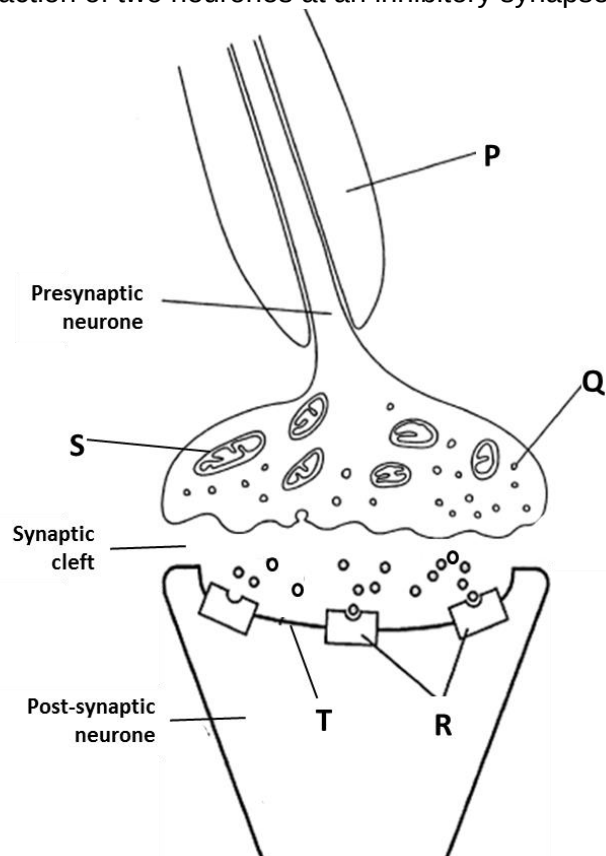
Question 29

Which of the following gives an accurate comparison between intracellular receptors and cell surface receptors?

	Intracellular receptors	Cell surface receptors
A.	May act as regulatory proteins and bind to DNA	May catalyse the phosphorylation of intracellular proteins
B.	Functions as the second messenger to activate other relay proteins	Binding of ligand always trigger the production of second messengers
C.	Ligands can be water-soluble or lipid-soluble	Ligands must be lipid-soluble
D.	Made up of only hydrophobic amino acids to allow the interaction with lipid-soluble ligands.	Made up of hydrophobic amino acids which interact with the phospholipids of the membrane

Question 30

The diagram shows the interaction of two neurones at an inhibitory synapse.



Which row correctly describes each of the labelled molecules?

	P	Q	R	S
A.	Contains a pool of Na^+ which enters the presynaptic neurone during saltatory conduction	Contains neurotransmitters which are triggered to undergo exocytosis by an influx of K^+	When bound, R opens to allow the <u>influx of Na^+</u> ions into the post-synaptic neurone	Produces ATP for the reabsorption of choline
B.	Insulates the axon to allow for more efficient nerve impulse transmission	Contains neurotransmitters which are triggered to undergo exocytosis by an influx of Ca^{2+}	When bound, R opens to allow the <u>influx of Na^+</u> ions into the post-synaptic neurone	Produces ATP for the active transport of Ca^{2+} into the synaptic knob
C.	Lipid in nature and hence prevent the movement of Na^+ and K^+ across the membrane	Contains neurotransmitters which are triggered to undergo exocytosis by an influx of Na^+	When bound by Q, R opens to allow the efflux of K^+ ions into the post-synaptic neurone	Produces ATP for the active transport of ions through the Na^+ - K^+ pump
D.	Made of a Schwann cell which facilitate the regeneration of injured cells	Contains neurotransmitters which are triggered to undergo exocytosis by an influx of Ca^{2+}	When bound, R opens to allow the <u>influx of Cl^-</u> ions into the post-synaptic neurone	Produces ATP for the active transport of Ca^{2+} out of the synaptic knob

Question 31

In one of the scientific journals that Charles Darwin published in 1839, he wrote:

"I have stated, that in the thirteen species of ground-finches [in the Galapagos Islands], a nearly perfect gradation may be traced, from a beak extraordinarily thick, to one so fine, that it may be compared to that of a warbler. I very much suspect that certain members of the series are confined to different islands; therefore, if the collection had been made on any one island, it would not have presented so perfect a gradation. It is clear, that if several islands have each their peculiar species of the same genera, when these are placed together, they will have a wide range of character."

What does the quote above illustrate?

- A. Convergent evolution, homologous structures, discontinuous variation
- B. Convergent evolution, analogous structures, continuous variation
- C. Divergent evolution, analogous structures, discontinuous variation
- D. Divergent evolution, homologous structures, continuous variation

Question 32

A plant population that reproduces annually exists on a remote oceanic island. Two flower-colour variants are present, red and blue. Flower colour is known to be a monogenic trait. The frequencies of the two flower colours were observed and it was noted that the proportion of plants with blue flowers steadily declined in each of ten generations.

From this data, it may reasonably be concluded that

- A. Migration of red-flowered plants into the population was the most likely cause of the observed change.
- B. Blue-flowered plants had a lower genetic fitness than the red-flowered plants.
- C. The increased frequency of red-flowered plants was a result of genetic drift.
- D. Mutation was producing more red-flowered plants in each generation.

Question 33

An enzyme consists of four domains (A–D), one of which contains the active site. The amino acid sequences of these four domains have been determined and compared in five related species. Which domain probably contains the active site?

Domain	Percentage of identical amino acids
A.	32%
B.	8%
C.	78%
D.	50%

Question 34

The steps involved in a method of production of human insulin by gene technology are listed below. The steps are not listed in the correct order.

Steps	Description
1	DNA coding for human insulin inserted into cut plasmid vector
2	genetically modified bacteria identified
3	mRNA for human insulin is harvested
4	plasmid vector inserted into bacterium
5	genetically modified bacteria cloned
6	DNA for human insulin cloned
7	cDNA coding for human insulin synthesised

Which of the following shows the correct sequence of steps?

A. 3 → 7 → 1 → 4 → 6 → 2 → 5

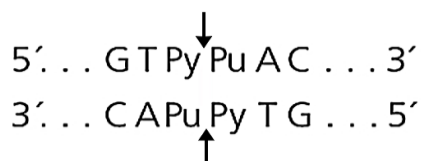
B. 7 → 4 → 1 → 6 → 2 → 5 → 3

C. 3 → 1 → 6 → 2 → 5 → 7 → 4

D. 7 → 1 → 4 → 6 → 2 → 3 → 5

Question 35

The restriction enzyme *Hind*III recognizes and cut at the following sequence:



where Py is any pyrimidine, and Pu is any purine.

What is the number of DNA fragments produced upon subjecting the following short length of linear DNA to *Hind*III digestion?

5' GATCGTCAACGTACGTCTACGTTACGTTGACATGAGCT 3'
 3' CTAGCAGTTGCATGCAGATGCAAGTGCAACTGTACTCGA 5'

A. 1

B. 2

C. 3

D. 4

Question 36

Green fluorescent protein (GFP) gene codes for GFP, a small protein that emits bright green fluorescence under UV light.

Genes coding for enzymes that act on substrates to produce fluorescent substances are often preferred over the GFP gene as markers in gene cloning.

Which of the following is/are the reason(s) why genes for enzymes that produce fluorescent substances are often preferred over the GFP gene in gene cloning?

1. GFP emits only green fluorescence, while enzymes that act on substrates to produce fluorescent substances can emit different colour fluorescents with different substrates.
2. Because enzymes can be reused, a few molecules of enzyme that produce fluorescent substances can produce a strong fluorescence as compared to the same number of GFP molecules.
3. Low expression of GFP gene may result in weak fluorescence that is difficult to detect, whereas low expression of genes for enzymes that produce fluorescent substances can still result in strong fluorescence by increasing the substrate concentration.

A. 2 and 3 only

B. 1 and 2 only

C. 1 and 3 only

D. 1, 2 and 3

Question 37

Which correctly describe the difference between genomic and cDNA libraries and the corresponding reason for the difference?

	Difference	Reason
A.	Genomic library may contain distant control elements and centromere while cDNA library contains only genes.	Due to mRNA splicing which removes introns and splice exons together.
B.	Genomic library is larger in size than cDNA library.	Due to genomic library containing total DNA extract from a cell while cDNA library containing the total mRNA extract from a particular cell type.
C.	Genomic library from an individual is always the same, while cDNA library may differ, depending on the cell type from which it is constructed.	Due to differential gene expression in the different cell types.
D.	Genes from genomic library are more suited than those from cDNA library for expression in bacterial cells.	Due to the action of restriction endonucleases in the construction of genomic library and that of reverse transcriptase in the construction of cDNA library.

Question 38

Which row correctly states the goal, benefits and concern arising from the Human Genome Project?

	Goal	Benefit	Concern
A.	To obtain the sequence of all the DNA nucleotides on all the human chromosomes. This would enable particular sequences to be patented so that advancements could be made in gene therapy.	Genetic screening for earlier detection of genetic predispositions to disease as increasingly detailed genome maps have aided researchers to seek out genes associated with dozens of genetic conditions.	Parents choosing embryos for implantation only after ante-natal tests for acceptable genes.
B.	To allow the various centres sequencing the DNA nucleotides on all the human chromosomes to share results on the databases. This would ensure that data were publicly available.	Assessment of risks posed to individuals caused by exposure to toxic agents such as radiation exposure, mutagenic chemicals and cancer causing toxins.	Genetic archaeologists identifying the earliest forms of genes to show evolutionary relationships.
C.	To develop techniques and technological tools for use in the sequencing project. This would lead to faster sequencing of sections of the human genome and improvements in mapping.	A better understanding of the human genome so that genetic counsellors can advise patients on the consequences and nature of the disorder, the probability of developing or transmitting it, and the options open to them in management and family planning.	An insurance company only giving cheap rates to people with genetic predispositions to fewer diseases.
D.	To reduce the gap between the rich and the poor by mapping the positions of all the genes on all the human chromosomes. This would help to identify the location of many genes that can cause disease.	Allows cytologists to develop diagnostic tests for defective genes.	The possibility of social discrimination that arises due to the knowledge of individual's genetic difference.

Question 39

Which of the statements about stem cells are true?

1. Embryonic stem cells are useful as they have the ability to differentiate into all cell types, except cells of the extraembryonic membranes.
2. Embryonic stem cells are more easily isolated compared to adult stem cells.
3. Umbilical cord blood stem cells have a higher developmental potential than skin stem cells.
4. Induced pluripotent stem cells undergo transdifferentiation to give rise to limited number of cell lineages.
5. Stem cells have the ability to self-renew without any stimulus.
6. Stem cells contain mitochondrial DNA.

A. 1, 2, 6

B. 1, 4, 6

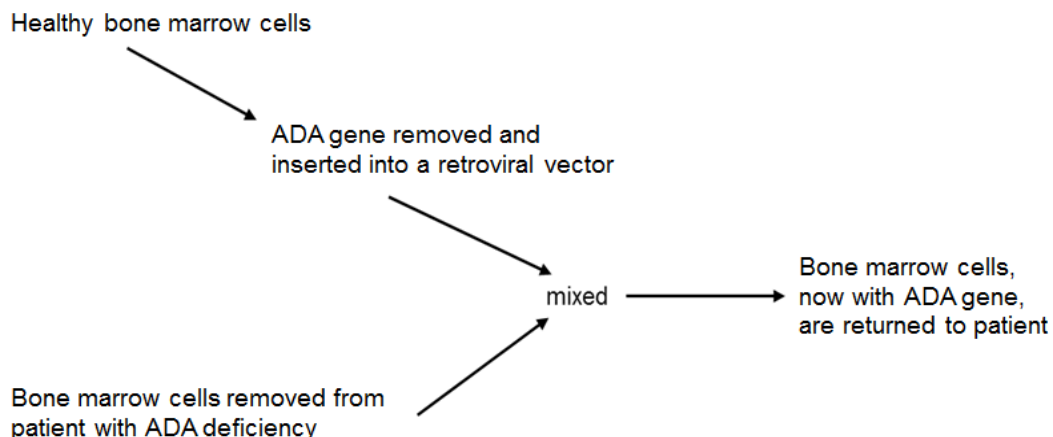
C. 2, 3, 4

D. 3, 5, 6

Question 40

Children with severe combined immunodeficiency disorder (SCID) cannot produce white blood cells that fight infections. This is because they do not have the functional gene to make the enzyme, adenosine deaminase (ADA).

Gene therapy can be used to treat SCID. The diagram shows a possible gene therapy approach.



Which of the following statements about the gene therapy approach are true?

1. Using this viral vector to introduce the normal ADA gene into the bone marrow cell is less efficient compared to using a non-viral vector.
2. This is an example of *ex-vivo* gene therapy
3. As the bone marrow cells returned to the patient were isolated from the patient, it will be genetically identical to the patient's cells.
4. Effects of the treatment will be permanent.
5. The patient may develop cancer.
6. The child of the patient will inherit the normal ADA gene.

A. 1, 2, 5

B. 2, 3, 6

C. 2, 4, 5

D. 3, 4, 6

END OF PAPER 1